SYED MOHAMMAD NASIR HUSSAIN

LEAD DATA SCIENTIST | GENERATIVE AI, RAG & APPLIED MACHINE LEARNING

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Data Scientist and AI professional with 8+ years of combined experience, including 7+ years in full-time roles and 1+ year of freelance work, delivering machine learning, Generative AI, NLP and advanced analytics solutions across healthcare, telecom and retail. Experienced in hybrid ML–LLM workflows and RAG pipelines using knowledge-base retrieval, semantic similarity, Claude-based classification, confidence-based routing and human-review fallbacks. Brings hands-on model development, technical ownership and production delivery, with measurable improvements in accuracy, turnaround time and manual effort.

TECHNICAL SKILLS

Programming & data: Python, SQL, Pandas, NumPy, SciPy, scikit-learn, Excel

Machine learning: XGBoost, LightGBM, Random Forest, SGD, TensorFlow, classification, regression, feature engineering, model validation

Generative AI & RAG: LLMs, Claude, LLM integration, Retrieval-Augmented Generation (RAG), AWS Bedrock Knowledge Bases, embeddings, semantic retrieval, vector similarity search, similarity-based matching, confidence-based routing, prompt engineering, human-in-the-loop review

NLP & analytics: Text classification, sentiment analysis, keyword extraction (YAKE), statistical analysis, exploratory data analysis, Tableau

AWS & cloud: Amazon Bedrock, S3, Lambda, Step Functions, EventBridge, SNS, SQS, SES, CloudWatch, CloudTrail, Athena, Redshift, SageMaker, Comprehend

PROFESSIONAL EXPERIENCE

Huron | Bengaluru, India *Sep 2021–Present*

Lead Data Scientist (Jan 2025–Present) | Associate Data Scientist, GenAI & Applied ML (Jan 2023–Dec 2024) | Senior Analyst (Sep 2021–Dec 2022)

- Own technical development and delivery across a portfolio of 7 AI/ML initiatives (4 deployed solutions and 3 proofs of concept), translating client requirements into solution designs, model workflows, validation plans and deployment-ready deliverables.
- Designed and delivered reusable hybrid ML + LLM categorization workflows for Clinical Supplies and Accounts Payable, combining deterministic lookup rules, LLM-based description enrichment, knowledge-base retrieval, confidence-based classification and manual-review pathways.
- Delivered six Clinical Supplies production models across three taxonomy levels and three orthopedic attribute domains; achieved 90%+ accuracy across six completed engagements for seven clients.

Pipeline optimization: Improved accuracy from 55% to 95%, reduced manual review from 67.5 to 7.5 hours per engagement, and shortened turnaround from 10 business days to 1–2 days. Projected annual savings were $101,467.50 across 15 engagements.

Completed engagements: Separately reported $17,100 in savings across six completed engagements, based on a 25-hour reduction in manual review per engagement. AP Vendor Categorization saved 50 hours per engagement and reported $22,800 in savings across four engagements.

- Lead model and pipeline execution, QA, troubleshooting, validation, production verification and technical documentation; collaborate with cross-functional teams and mentor colleagues through technical knowledge-sharing.

Ganit | Bengaluru, India *Mar 2021–May 2021*

Senior Data Analyst

- Developed a deep-learning model for retail pricing analysis using approximately 50,000 records; achieved an R^2 score of 0.90.
- Worked with a five-member team to translate model outputs into inputs for business pricing decisions.

Quadgen Wireless Solutions | Bengaluru, India *Sep 2017–Jul 2019*

Data Analyst

- Developed an ML-based anomaly-classification solution for cellular network behaviour across 10 base stations, achieving approximately 90% reported accuracy.
- Defined normal versus anomalous operating patterns, tuned model parameters and evaluated precision and recall to help reduce manual monitoring and support network operations.

ADDITIONAL EXPERIENCE

Freelance | Independent *Jul 2019–Mar 2021*

Data Science Trainer & Freelance Data Scientist

- Provided data science training and freelance analytics/model-development support across independent engagements.

SELECTED AI & MACHINE LEARNING PROJECTS

Clinical Supplies Categorization | Huron: Built a retrieval-assisted classification workflow combining exact-item lookup, Claude-based description enrichment, Knowledge Base retrieval and taxonomy-constrained classification. Added anatomical-site detection and consistency checks; retrieval uses the top 8 matches with 70% semantic and 30% text similarity.

AP Vendor Categorization | Huron: Built a GenAI + Knowledge Base workflow with curated vendor lookup, refund rules, vendor-description enrichment and staged Claude classification. Uses an 80% KB similarity threshold and adds GL account/department signals when Stage 1 Claude confidence is below 70%.

Rounding Vnext — Sentiment Analysis | Huron: Deployed AWS Comprehend sentiment analysis on hundreds of thousands of comments, achieving approximately 80% test accuracy; used YAKE to surface negative-feedback themes.

Revenue Cycle Authorization Prediction | Huron: Built an XGBoost POC on approximately 100,000 records to predict authorization requirements; approximately 97% reported accuracy.

IPEDS Graduation Rate Analysis | Huron: Modelled graduation-rate outcomes for 150 institutions across four academic years using Random Forest; benchmarked LightGBM and XGBoost (RMSE ±7.5 percentage points).

MRL Pricing Prediction | Ganit: Developed a deep-learning pricing model using approximately 50,000 retail records, achieving an R^2 score of 0.90.

Cellular Network Anomaly Detection | Quadgen: Developed an ML classifier for base-station anomalies across 10 sites, achieving approximately 90% reported accuracy.

RESEARCH PUBLICATION

Nasir, S. M., Pavate, A., & Yusuf, A. (2026). "RAG Pipeline Accuracy Benchmarking in Healthcare Supply Chain: Medical Item Classification Tasks." Karshi Multidisciplinary International Scientific Journal, 3(2), 153–164.

EDUCATION & AWARDS

B.E., Electronics & Communication Engineering | Aditya College of Engineering | 2013–2017 | CGPA: 7.5/10

Spot Awards — Quadgen Wireless Solutions and Huron | Rubik's Cube Award — Huron